Does the counterterm α_2 double count the Fingers-of-God dispersions?

The redshift-space counterterm against the sampled velocity dispersions σ_v, template by template and in the posterior

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1 Summary

The question. The forecast's one-loop model – the galaxy-clustering model one order beyond linear theory, defined in `notes/higher_order_bias` – carries two descriptions of nonlinear velocities along the line of sight. One is the Fingers-of-God blur: each bin's redshift distribution is convolved with a Gaussian whose width is set by a velocity dispersion σ_v, one per dispersion group (a set of bins sharing one σ_v; Table 1), sampled under a Gaussian prior. The other is a counterterm: a free term with a fixed scale shape, here $k^2 P_{\rm lin}$, that absorbs the small-scale physics the perturbative model cannot predict. The line-of-sight counterterm α_2 of the DESI 2024 full-shape convention [1] is

$$P_{\rm CT}(k,\mu) \;=\; \left(b_1 + f\mu^2\right) f\,\alpha_2\,\mu^2 k^2\,P_{\rm lin}(k), \tag{1}$$

with k the comoving wavenumber, μ the cosine of the angle to the line of sight, b_1 the bin's linear galaxy bias, $f = d\ln D/d\ln a$ the linear growth rate (D the linear growth factor, a the scale factor), $P_{\rm lin}$ the linear matter power spectrum, and α_2 in $h^{-2}\,{\rm Mpc}^2$, one value per group plus a free tilt (a straight line in redshift across the group's bins) in the fits that free it. The leading effect of the blur on the power spectrum is a damping $-\sigma_\chi^2 k_\parallel^2$ with $k_\parallel = k\mu$ and $\sigma_\chi = (1+z)\,\sigma_v/H(z)$ the dispersion in comoving distance: the same $k_\parallel^2$ shape as Eq. (1) (Section 3). So the question is whether the forecast pays twice for one effect: is α_2 degenerate with the σ_v already in the model?

Two fits. The study's reference is the fiducial bias model of `notes/higher_order_bias` on the pessimistic linear-bias prior, fitted with α_2 free and the exploration σ_v prior widths of Table 1 (the reference model). The *fit without* α_2 is the same model with α_2 and its tilts fixed at zero. The *paper model* – the configuration the paper yamls hold – is the fit without α_2 with all seven σ_v prior widths multiplied by $s = 6$.

The answer.

1. **At the template level, yes.** On every layout and at every cut, combinations of the seven σ_v templates reproduce 95.8–100% of each α_2 group template's information in the C^{-1} metric (C the band-power covariance), with medians 99.3–99.9%; the two templates are mirror images, so a positive α_2 looks like a smaller σ_v (Figure 1). Only the α_2 tilt columns are substantially new.

2. **In the posterior, the exploration σ_v priors break the degeneracy at small cuts but not at large ones.** In the reference model the largest correlation between an α_2 parameter and a σ_v grows, on the thin layouts, from $|\rho| = 0.39$–0.46 at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ to 0.81–0.88 at 0.2 (Figure 2).

3. **What α_2 costs.** Fixing α_2 and its tilts at zero raises $\mathrm{FoM}(w_0, w_a)$ of the reference model by a factor of 2.4–2.7 at 0.075 and 1.16–1.25 at 0.2 on the thin layouts; $\mathrm{FoM}(\Omega_m, 10^9 A_s)$ moves the same way. The gain comes entirely from the σ_v priors: with flat σ_v priors, fixing α_2 returns 0.83–1.02 of the reference model's FoM at every cut from 0.075 on.

4. **What dropping α_2 risks.** If the truth holds a DESI-prior-sized $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ in every group, the fit without α_2 is shifted in (w_0, w_a), and σ_v on the exploration priors absorbs little of it: the shift falls by 4% and 15% at 0.075 and by 34% and 39% at 0.2 on the two thin layouts. With flat σ_v priors those shifts fall by 87–98%. The wide layouts barely react either way. These shifts estimate how safe an MCMC with the model would be on real data; the paper itself is Fisher-only.

5. **A wider σ_v prior trades the gain for the absorption.** Multiplying all seven σ_v prior widths by a common factor s, the fit without α_2 absorbs the $12.5\,h^{-2}\,\mathrm{Mpc}^2$ truth (every shift below $\sigma/3$) at $s = 1.35$ and 1 at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ on the thin layouts, where it still keeps 2.3 and 2.7 times the reference model's FoM; at 0.1–0.2 it absorbs only at $s = 3.6$–6.3, where the FoM ratio is 0.93–1.16 and the gain is gone (Figure 3). Widening one sample's dispersions alone never suffices on the thin layouts at 0.1 and above.

6. **The paper adopts $s = 6$.** Evaluated directly, the largest shift of the paper model under $\alpha_2 = \pm 12.5\,h^{-2}\,\mathrm{Mpc}^2$ is 0.327σ ($\Delta z = 0.02$ at $k_{\max} = 0.1\,h/\mathrm{Mpc}$), below $\sigma/3$ at every layout and cut; the interpolated passing threshold reaches 6.3 on that one point, so the margin there is small. The width was chosen so that dropping α_2 does not inflate the figure of merit: on the thin layouts the paper model's $\mathrm{FoM}(w_0, w_a)$ is at most 1.04 times the reference model's at 0.1 and 0.89–1.00 of it from 0.125 on, against the factor 1.7–2.1 at 0.1 of item 3's fixed fit at $s = 1$ (Section 6 has the full ratios, including the wide-layout exceptions).

7. **Sampling σ_v can outgrow the pair list.** The data vector's cross pairs are fixed by an overlap rule at the fiducial σ_v ($O_{ij} \geq 10^{-6}$, Section 7). Chains over the exploration priors stay inside the list: the nearest unlisted pair crosses the threshold at +2.1 prior widths on $\Delta d_{\mathrm{com}} = 62$ Mpc. The paper model's widened priors do not: the crossing sits at 0.34 and 0.52 of the widened width on the thin layouts, and at their $+2\sigma$ edge pairs of BGS bins two apart overlap at 7.1×10^{-2} and 2.5×10^{-2}, far above the threshold. The lists selected at those larger dispersions are forecast in `notes/cross_correlations` (its σ_v axis): the added pairs raise $\mathrm{FoM}(w_0, w_a)$ by up to a factor of about 1.6 on the three thinner layouts, by separating each group's free bias level and slope from the growth history rather than through line-of-sight information of their own.

Here w_0 and w_a parameterize the dark-energy equation of state $w(a) = w_0 + w_a(1 - a)$; Ω_m is the matter density today in units of the critical density; A_s is the amplitude of the primordial curvature power spectrum, sampled as $10^9 A_s$; and $\mathrm{FoM}(w_0, w_a) = [\det \mathrm{Cov}(w_0, w_a)]^{-1/2}$ is

the inverse area of the marginalized (w_0, w_a) error ellipse – marginalized meaning every other parameter is left free, so their uncertainty is included – and likewise FoM$(\Omega_m, 10^9 A_s)$: a larger FoM is a tighter constraint. In short: one $k_\parallel^2$ effect, counted twice at the template level; the posterior keeps the two apart only as long as the σ_v priors do, and the whole dark-energy gain of dropping α_2 is exactly as strong as those priors. The paper model drops α_2 and widens the σ_v priors by the factor that surrenders that gain, so that a true DESI-size α_2 is absorbed.

The setup. The data are the DESI angular clustering band powers C_ℓ^{gg} alone – the angular power spectrum of galaxy counts in tomographic redshift bins, averaged in bands of multipole ℓ – computed without the Limber approximation, which ties each multipole to a single wavenumber and fails for thin bins [7], on four bin layouts: $\Delta d_{\rm com} = 62$ Mpc (equal comoving width) and $\Delta z = 0.02$, 0.05 and 0.2 (fixed redshift width); the first two are the thin layouts. The scan runs over $k_{\rm max} = 0.05$ to $0.2\,h$/Mpc, each the $z = 0$ value of a per-bin cut at $k_{\rm max}/D(\bar{z}_i)$, with $\bar{z}_i$ the mean redshift of bin i. Every result is a Fisher forecast: the data vector is linearized at the fiducial point, and the posterior is approximated by a Gaussian whose inverse covariance is the Fisher matrix of the appendix; `notes/fisher_machinery` builds that machinery step by step. Every number and figure of this note comes from section 11 of `EXAMPLE_EVALUATE_FISHER_MODEL.ipynb`; the per-cut values are in the four figure folders' `data.json`.

2 Fiducial model and parameter values

This note was built at one fiducial point; the values are written out here so the note does not depend on configuration files that may change.

Cosmology. Flat $w_0 w_a$CDM: $10^9 A_s = 2.1$, $n_s = 0.96605$, $H_0 = 67.32$ km/s/Mpc, $\Omega_b = 0.04$, $\Omega_m = 0.3$, $w_0 = -1$, $w_a = 0$ (sampled as $w_0 + w_a = -1$), $\sum m_\nu = 0.06$ eV and $\tau = 0.0697186$ (both fixed). The derived $\sigma_8 = 0.8278$. The Fisher matrices vary $10^9 A_s$, n_s, H_0, Ω_b, Ω_m, w_0 and w_a, all with flat priors.

Layouts. Every layout uses the same 53 band powers of width $\Delta\ell = 10$ over $\ell = 15$ to 544 per spectrum.

layout	lens bins	cross pairs	C_ℓ^{gg} entries
$\Delta d_{\rm com} = 62$ Mpc	76	107	10494
$\Delta z = 0.02$	90	133	12614
$\Delta z = 0.05$	36	52	5459
$\Delta z = 0.2$	10	13	2014

Entries count every spectrum of the data vector (the 15 cosmic-shear spectra of the 5 LSST Y1 source bins included) times the 53 bands; this study scores the DESI C_ℓ^{gg} block alone.

Linear bias. Bin i of group g has $b_{1,i} = B_g / D(\bar{z}_i)$ at the fiducial, with D the linear growth factor read off the linear $P(k)$ ratio at $k = 0.005$/Mpc and the group amplitudes $B_{\rm BGS} = 1.34$, $B_{\rm LRG1} = B_{\rm LRG2} = B_{\rm LRG3} = 1.7$, $B_{\rm ELG1} = B_{\rm ELG2} = 0.84$. The linear-bias prior is the pessimistic one of `notes/correlated_bias_prior_v2`: per-bin residuals around each group's free amplitude and free straight line in redshift, with width $\sigma_g = 0.03$ in every group and

correlation length $L = 0.05$ in redshift; the Fisher uses the projected precision, that prior's inverse covariance with the amplitude and line directions removed, so only the wiggles around each group's line are charged and the amplitude and the line stay flat.

Higher-order terms. The bias model is the fiducial model: per group, offsets Δ_2 and Δ_s of b_2 and b_{s^2} from their calibrations (b_2 from the Lazeyras et al. fit [6], $b_{s^2} = -(4/7)(b_1 - 1)$, $b_3 = 0$), the counterterms α_0 and α_2, and the stochastic amplitudes SN_0 and SN_2, with a free straight line in redshift (a tilt) on Δ_2, Δ_s, α_0 and α_2. Every offset, counterterm, stochastic amplitude and tilt is zero at the fiducial. Gaussian prior widths per group: Δ_2: 2.0; Δ_s: 1.0; α_0 and α_2: $0.5/k_{\mathrm{max}}^2$ at each cut ($12.5\,h^{-2}\,\mathrm{Mpc}^2$ at $k_{\mathrm{max}} = 0.2\,h/\mathrm{Mpc}$); SN_0: 2 (in units of the bin's Poisson shot noise $1/\bar{n}_i$); SN_2: 5; tilts flat. The paper model fixes α_2 and its tilts at zero in every bin.

Velocity dispersions. Seven dispersion groups (BGS split at $\bar{z} = 0.255$), each sampled on its group's leader bin – one designated bin carries the sampled parameter, and every bin of the group reads its value – with the fiducials and prior widths of Table 1: the reference model and the width scan's $s = 1$ use the exploration widths; the paper model uses six times those widths.

Scale cuts. The non-Limber ultraviolet-fraction criterion (arXiv:2603.13031): a band is kept while $|C_{\mathrm{UV}}|/(|C_{\mathrm{UV}}| + |C_{\mathrm{IR}}|) < 0.05$, with C_{UV} and C_{IR} the parts of the band's non-Limber integral from wavenumbers above and below the split at $k_{\mathrm{max}}/D(\bar{z}_i)$ per bin, intersected with the shipped mask, the dataset's own fixed cut at $k = 0.1185/\mathrm{Mpc}$ in every bin. The cuts are computed on the paper model's fiducial (b_2 and b_{s^2} on their calibrations, every other higher-order term zero), at the final derivative tier (the settings of the derivatives below), and cached in `chains/model_scale_cuts_final.npz`.

Covariance. The Gaussian band-power covariance, built at 14000 deg^2 on the one-loop fiducial (b_2, b_{s^2} on their calibrations, massless neutrinos, the FoG blur at the fiducial σ_v); every Fisher matrix rescales its inverse by $12300/14000 = 0.879$.

Derivatives. The final derivative tier: five-point stencils at relative step $h = 0.02$ (α_2, SN_0 and SN_2 enter as exact linear templates), likelihood settings accuracyboost 3, nonlimber_accuracyboost 64, integration_accuracy 1, kmax_boltzmann 20/Mpc, and CAMB at AccuracyBoost 2, k_per_logint 30 with the halofit tolerance patch. The base, built on the paper yamls `EXAMPLE_EVALUATE1-4.yaml`, is cached in `chains/model_derivatives_<layout>_final.npz` and `chains/model_templates_<layout>_final.npz`.

3 Why the question is natural

Redshift is blurred distance. The forecast places each galaxy at its measured redshift. Inside groups and clusters, galaxies orbit with random speeds of several hundred km/s, so a galaxy's measured redshift carries a Doppler shift on top of its distance: in a redshift map a cluster stretches into a finger pointing at the observer, the Fingers of God [3]. A line-of-sight

velocity v at redshift z displaces the galaxy by

$$\delta\chi \;=\; \frac{(1+z)\,v}{H(z)}, \tag{2}$$

with χ the comoving distance and $H(z)$ the expansion rate. A dispersion σ_v of these velocities blurs distances by $\sigma_\chi = (1+z)\,\sigma_v/H(z)$: for the groups here, 5–6 Mpc, about $4\,h^{-1}$Mpc (`notes/cross_correlations`, which derives the convolution and its sizes).

Blurring the window equals blurring the field. The likelihood implements the blur by convolving each bin's redshift distribution $n_i(z)$ with a Gaussian of width $(1+\bar{z}_i)\,\sigma_v/c$ in redshift [5]. That window-side convolution is the same operation as damping the three-dimensional field, as long as the velocities are a random scatter uncorrelated with the density. Write the observed galaxy field as the true one with each galaxy moved by $\delta\chi$ drawn from a distribution p: along one line of sight, $\delta_g^{\rm obs}(\chi_{\rm obs}) = \int d\chi\, p(\chi_{\rm obs}-\chi)\,\delta_g(\chi)$. A bin's map integrates the observed field over its radial window W_i:

$$\begin{aligned}\int d\chi_{\rm obs}\, W_i(\chi_{\rm obs})\,\delta_g^{\rm obs}(\chi_{\rm obs}) \;&=\; \int d\chi \Big[\int d\chi_{\rm obs}\, W_i(\chi_{\rm obs})\,p(\chi_{\rm obs}-\chi)\Big]\,\delta_g(\chi)\\ &=\; \int d\chi\,\big(W_i{*}p\big)(\chi)\,\delta_g(\chi)\,:\end{aligned} \tag{3}$$

swapping the order of integration moves the convolution from the field onto the window. Blurring every bin's $n(z)$ and keeping the true field is therefore the same map as keeping sharp bins and blurring the field, which is how the likelihood's σ_v reach the spectra.

The damping and its leading order. On the field side, a convolution along the line of sight multiplies each Fourier mode by the distribution's transform: for a Gaussian p of width σ_χ, the density picks up $\exp(-k_\parallel^2\sigma_\chi^2/2)$ and the power spectrum $\exp(-k_\parallel^2\sigma_\chi^2)$, with $k_\parallel = k\mu$ the line-of-sight wavenumber. Applied to the linear redshift-space power $(b_1+f\mu^2)^2 P_{\rm lin}$ – the Kaiser form [4], in which the coherent infall toward overdensities adds $f\mu^2$ to the bias b_1 and so boosts the clustering along the line of sight – to leading order in $k_\parallel^2\sigma_\chi^2$,

$$\big(b_1+f\mu^2\big)^2 P_{\rm lin}\,e^{-k_\parallel^2\sigma_\chi^2} \;\simeq\; \big(b_1+f\mu^2\big)^2 P_{\rm lin} \;-\; \sigma_\chi^2\,k_\parallel^2\,\big(b_1+f\mu^2\big)^2 P_{\rm lin}. \tag{4}$$

The counterterm, Eq. (1), is $f\,\alpha_2\,k_\parallel^2\,(b_1+f\mu^2)\,P_{\rm lin}$. Both are $k_\parallel^2$ times the linear power; their μ structures differ by exactly one factor of $(b_1+f\mu^2)$, and a band power averages much of that difference away. The signs are opposite where both amplitudes are positive: a positive α_2 raises the spectrum where a larger σ_v lowers it, so a positive α_2 looks like a *smaller* σ_v. The sizes are comparable too: $\sigma_\chi \approx 4\,h^{-1}$Mpc gives $\sigma_\chi^2 \approx 16\,h^{-2}\,{\rm Mpc}^2$, the order of the DESI prior width $12.5\,h^{-2}\,{\rm Mpc}^2$ of α_2 at $k_{\rm max} = 0.2\,h/{\rm Mpc}$.

Why both exist. In a three-dimensional full-shape analysis – a fit to the galaxy power spectrum $P(k,\mu)$ itself rather than to its projection on the sky – there is no radial window to convolve, so there is no sampled σ_v: the counterterm α_2 is that analysis's effective description of small-scale velocities, Fingers of God included [1, 2]. This forecast works on projected spectra, where the window-side convolution is natural and σ_v already carries the effect; α_2 was added

with the rest of the DESI counterterm set. A setup with α_0 only – the density counterterm, which multiplies $k^2 P_{\rm mm}$ with no μ – is the natural alternative, and this note measures what each choice costs.

The seven dispersions and their priors. Each bin's σ_v is its dispersion group's value (Table 1): the measured dispersion curve of the source forecast's simulated galaxies has a step inside BGS, so BGS takes two values, split at $\bar{z} = 0.255$, giving seven dispersion groups for six bias groups. Each prior is a Gaussian centred on the measured value and truncated at zero; its provenance is the section "Where the velocity dispersions come from" of `notes/cross_correlations`. The exploration widths, five times the median per-bin error bar of each group's measurement, are the priors of the reference model (the exploration files `likelihood/params_desi*.yaml`) and the $s = 1$ point of the width scan of Section 6; the paper model widens every width by $s = 6$ (the paper files `likelihood/params_desi*_bias_pessimistic.yaml`). The same seven values and widths serve every layout. At $s = 6$ the truncation matters for BGS0: its prior is a Gaussian of center 371 km/s and width 870 km/s cut at zero, which sits only 0.426 widths below the center, so 33.5% of the Gaussian mass is removed and the truncated prior has mean 847 km/s (about 850) and median 748. The Fisher matrices treat every σ_v prior as an untruncated Gaussian of the quoted width centred on the fiducial, so near zero the Fisher's prior is slightly wider than the real one; an MCMC with these files would see the truncation.

dispersion group	σ_v [km/s]	exploration width [km/s]	paper width ($\times 6$) [km/s]
BGS, $\bar{z}_i < 0.255$	371	145	870
BGS, $\bar{z}_i \geq 0.255$	358	105	630
LRG1	357	70	420
LRG2	348	50	300
LRG3	343	50	300
ELG1	304	40	240
ELG2	289	30	180

Table 1: The fiducial velocity dispersion of each group and the two widths $\Delta\sigma_{v,g}$ of its Gaussian prior: the exploration width (the reference model and $s = 1$ of the width scan) and the paper model's width, six times larger.

4 Templates, and how to measure redundancy

A parameter's template. A parameter p's template is the derivative of the data vector, $\partial C_\ell / \partial p$ over every kept band of every spectrum. To first order it is the fingerprint p leaves on the data: the fit can distinguish two parameters only as far as their templates differ. Each parameter is measured in units of its prior width Δ_p (written Δ, never σ, since σ_v is itself a dispersion): here $\Delta\alpha_2 = 0.5/k_{\rm max}^2$, the reference model's prior width, quoted at $k_{\rm max} = 0.2\,h/{\rm Mpc}$ where it is $12.5\,h^{-2}\,{\rm Mpc}^2$, and $\Delta\sigma_{v,g}$ each dispersion group's exploration width (Table 1). Divided by each band's error $\sigma_\ell = \sqrt{\mathsf{C}_{ii}}$, the scaled template $\sigma_\ell^{-1}\,\partial C_\ell / \partial (p/\Delta_p)$ reads directly as how many error bars the parameter can move each band while staying within its prior; Figure 1 draws the templates this way, following the scaled-template presentation

of Obied et al. [8]. The α_2 templates here are group templates: the response to the group's value, all bins of the group moving together (and, for the tilts, moving along a straight line in redshift).

The leftover, measured in the data's own error bars. Can an α_2 group template be rebuilt as a weighted sum of the seven σ_v templates? The leftover of the best weights c_g is

$$r_\ell \;=\; \frac{\partial C_\ell}{\partial(\alpha_2/\Delta\alpha_2)} \;-\; \sum_g c_g \,\frac{\partial C_\ell}{\partial(\sigma_{v,g}/\Delta\sigma_{v,g})}, \tag{5}$$

the sum running over the seven dispersion groups. It should not be measured band by band: a mismatch of $0.1\sigma_\ell$ on a precise band is visible where the same mismatch on a noisy band is not. The natural norm is the data's own, $r^{\mathsf{T}}\mathsf{C}^{-1}r$ with r the vector of the r_ℓ, the χ^2 the leftover would contribute, and the weights c_g minimize it. Writing a for the scaled α_2 template, the first term of Eq. (5), the redundancy fraction is the share of the template's information that the σ_v templates reproduce,

$$f \;=\; 1 \;-\; \frac{\min_c \, r^{\mathsf{T}}\,\mathsf{C}^{-1}\,r}{a^{\mathsf{T}}\,\mathsf{C}^{-1}\,a}. \tag{6}$$

$f = 1$ means the data cannot tell α_2 from that combination of dispersions at all. (The letter f follows the figure axis; the growth rate is written f only inside Eqs. (1) and (4).)

A three-band toy. Take three bands with unit error bars and two templates, $a = (1, 2, 3)$ and $s = (2, 4, 6)$. Then $a = s/2$ exactly: the leftover is zero and $f = 1$; whatever signal a could produce, $s/2$ produces identically, and the data cannot separate the two parameters. Now take $s = (2, 4, 5)$, which differs from $2a$ only on the third band. The best single weight is $c = 5/9$, the leftover is $r = (-1, -2, 2)/9$ with $r^{\mathsf{T}}r = 1/9$, and $f = 1 - (1/9)/14 = 0.992$. Whether that mismatch matters is decided by the third band's error bar. If the third band is noisy, the fit matches the first two bands almost exactly and hides the whole mismatch inside the third band's error: the leftover χ^2 falls toward zero and $f \to 1$. If the third band is precise, the fit is pinned there ($c \to 3/5$) and the mismatch is pushed onto the first two bands as a visible leftover. Either way, a mismatch is only as detectable as the error bars of the bands where it must appear – which is what the C^{-1} weighting in Eq. (6) counts.

One parameter against one: $f = \rho^2$. With one α_2 amplitude, one σ_v and nothing else free (templates a and s), the two-parameter Fisher matrix is built from $a^{\mathsf{T}}\mathsf{C}^{-1}a$, $a^{\mathsf{T}}\mathsf{C}^{-1}s$ and $s^{\mathsf{T}}\mathsf{C}^{-1}s$, and the correlation of its inverse is $\rho = -\,a^{\mathsf{T}}\mathsf{C}^{-1}s\,/\sqrt{(a^{\mathsf{T}}\mathsf{C}^{-1}a)(s^{\mathsf{T}}\mathsf{C}^{-1}s)}$. The best single weight in Eq. (5) is $c = a^{\mathsf{T}}\mathsf{C}^{-1}s/(s^{\mathsf{T}}\mathsf{C}^{-1}s)$, which leaves $\min_c r^{\mathsf{T}}\mathsf{C}^{-1}r = a^{\mathsf{T}}\mathsf{C}^{-1}a - (a^{\mathsf{T}}\mathsf{C}^{-1}s)^2/(s^{\mathsf{T}}\mathsf{C}^{-1}s)$; dividing by $a^{\mathsf{T}}\mathsf{C}^{-1}a$ as in Eq. (6) gives

$$f \;=\; \frac{(a^{\mathsf{T}}\mathsf{C}^{-1}s)^2}{(a^{\mathsf{T}}\mathsf{C}^{-1}a)(s^{\mathsf{T}}\mathsf{C}^{-1}s)} \;=\; \rho^2.$$

Mirror-image templates ($a^{\mathsf{T}}\mathsf{C}^{-1}s < 0$) give a *positive* posterior correlation: the direction the data cannot see has α_2 and σ_v rising together, each undoing the other's change of the spectra. The posterior correlation of the full model is a different number from f: it marginalizes every other parameter – cosmology, biases, the other counterterms – which can soak up part of either template, and it includes the priors, which hold parameters the data would let drift. That is why f can sit at 0.99 while $|\rho|$ ranges from 0.002 to 0.89 across cuts and layouts.

Many against many: canonical correlations. For the full sets – six α_2 group columns (plus tilts) against seven σ_v columns – the squared canonical correlations generalize ρ^2: the first is the largest f any combination of α_2 columns reaches against any combination of σ_v columns, the second repeats the question in the directions orthogonal to that first pair, and so on. Geometrically they are the squared cosines of the principal angles between the two template subspaces in the C^{-1} metric, the per-direction version of Eq. (6).

5 Results

The templates are mirror images. Figure 1 draws the α_2 and σ_v templates of three $\Delta d_{\rm com} = 62$ Mpc spectra, each scaled by its prior width. The two templates track each other band by band with opposite signs, at a few tenths of σ_ℓ per band. The bottom row is the leftover r_ℓ of Eq. (5): after the best combination of the seven σ_v templates, fitted over the whole data vector, $|r_\ell|/\sigma_\ell$ stays at or below 0.04 per band in the BGS panel and below 0.013 in the LRG panels at the $0.2\,h$/Mpc cut, and below 0.005 everywhere at 0.075: templates of a few tenths of an error bar cancel to a few thousandths.

The redundancy against the cut. Over all four layouts and every cut of the scan, f of the six α_2 group columns lies between 0.958 and 1.000, with per-cut medians of 0.993–0.999; the squared canonical correlations of the six-column α_2 set with the seven-column σ_v set lie in the same range, 0.958–1.000. The top panel of Figure 2 draws $100\,(1-f)$ for the least redundant group: even there, at most 4.3% of the template's information is new. The α_2 *tilt* columns are much less redundant – their per-cut medians lie between 0.02 and 0.24 at every cut except the smallest cut on $\Delta d_{\rm com} = 62$ Mpc, where only three tilt columns keep any band – because a straight line in redshift across a group is a shape no combination of seven group-constant dispersions reproduces. At the template level, then, the α_2 group values duplicate information the σ_v already carry; the tilts are the genuinely new freedom.

The posterior of the reference model. The second panel of Figure 2 shows the largest $|\rho(\alpha_2, \sigma_v)|$ in the reference model's posterior. On the thin layouts it grows from 0.46 ($\Delta d_{\rm com} =$ 62 Mpc) and 0.39 ($\Delta z = 0.02$) at $k_{\rm max} = 0.075\,h$/Mpc to 0.81 and 0.88 at 0.2, peaking at 0.84 and 0.89 at intermediate cuts; on $\Delta z = 0.05$ it stays below 0.37, and on $\Delta z = 0.2$ below 0.08. Wherever the correlation is appreciable, the pair is LRG3's α_2 with LRG3's own dispersion, and its sign is positive, as mirror-image templates give (Section 4). The correlations stay below one because the σ_v priors hold their end of the degenerate direction: with α_2 free, every σ_v posterior keeps 0.60–1.00 of its prior width, so the dispersions stay prior-dominated while α_2 takes the $k_\parallel^2$ information. With α_2 and its tilts fixed, the data begin to measure the dispersions: on the thin layouts the best-measured σ_v tightens to 0.25–0.75 of its prior width at cuts of 0.075 and above.

What α_2 costs. The third panel of Figure 2 draws $\mathrm{FoM}_{\alpha_2=0}/\mathrm{FoM}$: the reference model with α_2 and its tilts fixed at zero, over the reference model. On the thin layouts the fixed fit has 2.40 and 2.74 times the $\mathrm{FoM}(w_0, w_a)$ at $k_{\rm max} = 0.075\,h$/Mpc, 1.74 and 2.11 at 0.1, and 1.16 and 1.25 at 0.2; $\mathrm{FoM}(\Omega_m, 10^9 A_s)$ gains similar factors (2.25 and 2.21 at 0.075, 1.18 and 1.25 at 0.2). On $\Delta z = 0.05$ the gain is 1.05–1.17, and on $\Delta z = 0.2$ at most 1.07 at every cut except

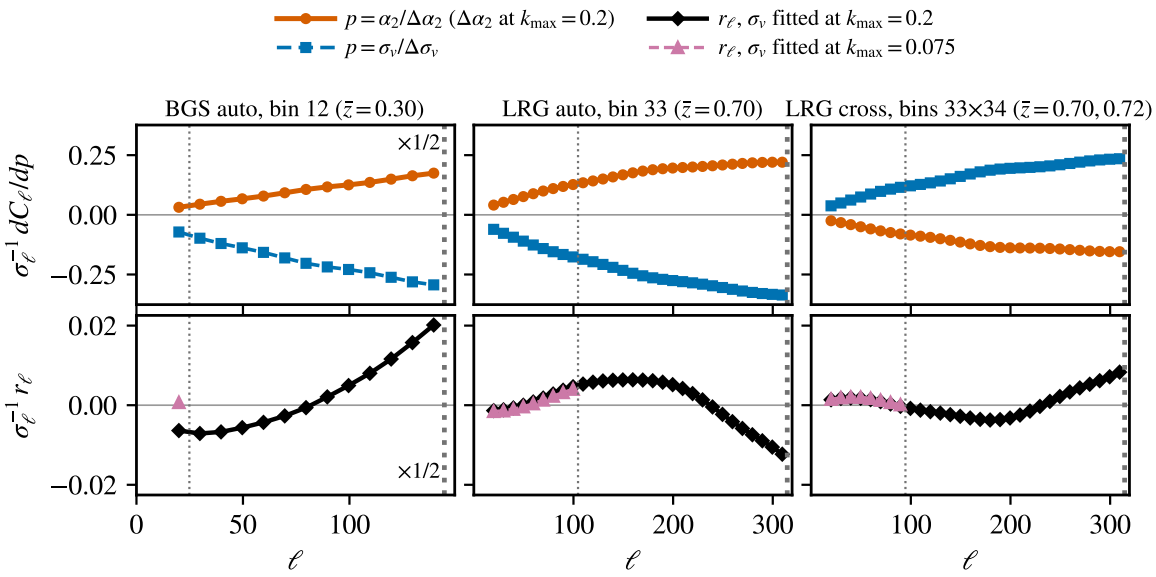


Figure 1: The α_2 and σ_v templates of three $\Delta d_{\rm com} = 62$ Mpc spectra – a BGS auto spectrum ($\bar{z} = 0.30$), an LRG auto spectrum ($\bar{z} = 0.70$) and that bin's cross spectrum with the next LRG bin – in units of each band's error $\sigma_\ell = \sqrt{\mathsf{C}_{ii}}$, on the bands the shipped mask keeps; the BGS column is divided by the round factor written in its panels, and each row shares one y axis. Top row: the scaled templates $\sigma_\ell^{-1}\, \partial C_\ell/\partial p$, each parameter in units of its prior width: $p = \alpha_2/\Delta\alpha_2$ of the bin's group, with $\Delta\alpha_2$ the prior width at $k_{\max} = 0.2\,h/\mathrm{Mpc}$ (vermillion solid, circles), and $p = \sigma_v/\Delta\sigma_v$ of the bin's dispersion (blue dashed, squares). Bottom row: $\sigma_\ell^{-1} r_\ell$, the leftover of the α_2 template after its best combination of the seven σ_v templates, Eq. (5), fitted over the whole data vector in the C^{-1} metric on the bands a cut keeps and drawn on those bands, for $k_{\max} = 0.2$ (black, diamonds) and $0.075\,h/\mathrm{Mpc}$ (pink dashed, triangles). Dotted vertical lines: where the shown spectrum's kept bands end at 0.075 (thin) and at 0.2 (thick). $k_{\max}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

0.125, where it reaches 1.19. The dashed curves isolate where the gain comes from: with the σ_v priors removed (flat), fixing α_2 returns 0.83–1.02 of the reference model's FoM at every cut from 0.075 on – no gain at all. Fixing α_2 therefore does not add information of its own; it hands the $k_\parallel^2$ direction to the dispersions, and the constraint the fixed fit gains is exactly the constraint the σ_v priors supply.

What dropping α_2 risks. The absorption test fits the fit without α_2 – the reference model with α_2 and its tilts fixed at zero – to a truth with $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ in every group, the DESI prior width at $k_{\max} = 0.2\,h/\mathrm{Mpc}$. The shift is a linear-response estimate – the parameter displacement predicted by the same linearization that builds the Fisher matrix, not a refit – of how safe an MCMC with the reduced model would be on real data (the paper itself is Fisher-only), and the +12.5 and −12.5 truths shift equally in this response (appendix "Method and checks"). Shifts are quoted in ellipse units: the (w_0, w_a) displacement measured against the fit's marginalized one-sigma ellipse in the displacement's direction, so one unit moves the best

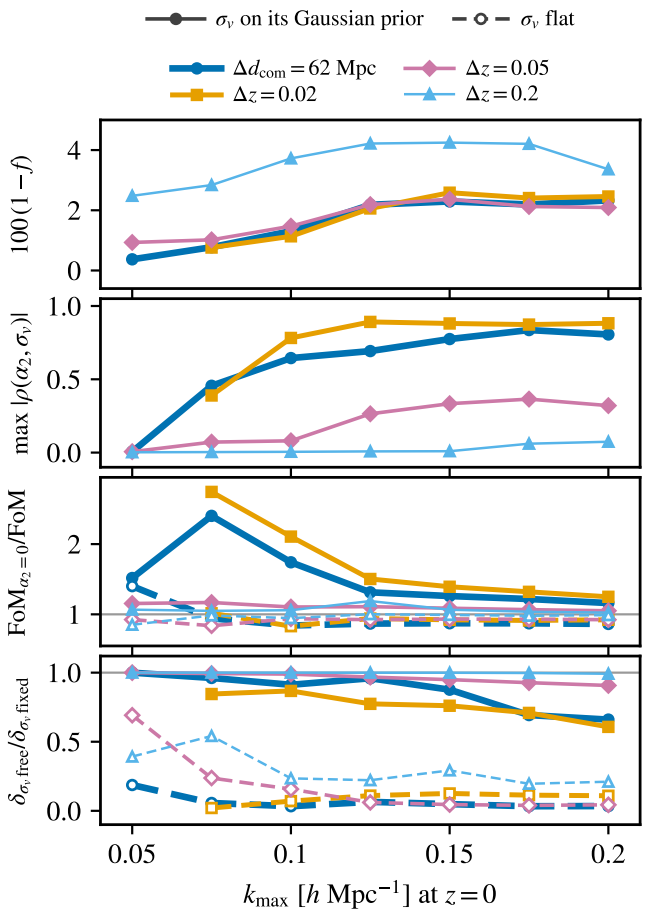


Figure 2: How far the counterterm α_2 and the Fingers-of-God dispersions σ_v describe the same band powers, against the scale cut, per layout (color, line width and marker; legend). Top panel: $100\,(1-f)$, the share of an α_2 group template's information in the C^{-1} metric that no combination of the seven σ_v templates reproduces, for the group where that share is largest; this panel is built from the templates and the covariance alone and does not depend on any prior. Second panel: the largest $|\rho|$ between an α_2 parameter and a σ_v in the posterior of the reference model, the fiducial model on the pessimistic linear-bias prior with α_2 free and the exploration σ_v widths. Third panel: $\mathrm{FoM}_{\alpha_2=0}/\mathrm{FoM}$, the $\mathrm{FoM}(w_0, w_a)$ with α_2 and its tilts fixed at zero over the reference model's, with σ_v on its Gaussian priors (solid, filled markers) or flat (dashed, open markers). Bottom panel: $\delta_{\sigma_v\,\mathrm{free}}/\delta_{\sigma_v\,\mathrm{fixed}}$, with δ the (w_0, w_a) shift of the fit without α_2 (the reference model with α_2 and its tilts fixed at zero) fitted to a truth with $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ in every group: the shift with σ_v free over the shift with σ_v fixed at its fiducial, with the same two line styles. Gray: one. $\Delta z = 0.02$ keeps no band at $0.05\,h/\mathrm{Mpc}$. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

fit onto its own one-sigma contour. If σ_v could stand in for α_2 in practice, freeing it would absorb the shift. On the exploration priors it barely does: the (w_0, w_a) shift falls from the fixed-σ_v value by 4% at 0.075 (0.328 → 0.315 ellipse units) and 34% at 0.2 (1.63 → 1.08) on $\Delta d_{\rm com} = 62$ Mpc, and by 15% (0.304 → 0.257) and 39% (2.54 → 1.54) on $\Delta z = 0.02$; on $\Delta z = 0.05$ by at most 9%, and on $\Delta z = 0.2$ the shift never exceeds 0.06σ to begin with. With flat σ_v priors the same shifts, at cuts of 0.075 and above, fall by 87–98% (bottom panel of Figure 2): the dispersions *can* mimic the counterterm, but only by moving far outside their priors. Fitting the $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ truth with the σ_v templates alone makes this concrete: on $\Delta d_{\rm com} = 62$ Mpc at 0.2 the truth carries $\Delta\chi^2 = 162$, of which free dispersions leave 1.6 – but the mimicking shift is -2.5 prior widths (-75 km/s, always the ELG2 dispersion, whose prior is tightest), and with the Gaussian priors the total left, data residual plus prior penalty, is 14. Across the thin layouts at cuts of 0.075 and above, the mimicking shift is 1.9–2.6 prior widths (56–78 km/s). The shifts scale linearly with the truth: a truth at one α_2 prior width ($0.5/k_{\rm max}^2$, $88.9\,h^{-2}\,\mathrm{Mpc}^2$ at 0.075) gives 7.1 times the $12.5\,h^{-2}\,\mathrm{Mpc}^2$ shifts there.

6 How wide must the σ_v priors be?

The knob. The absorption test leaves a natural follow-up question: if σ_v keeps a Gaussian prior, how wide must that prior be for the dispersions to absorb a DESI-size α_2? The two preceding results frame the answer before any scan. Mimicking $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ costs the dispersions 1.9–2.6 prior widths, so widening the priors by a factor of a few makes that move cheap and the shift disappears. But the FoM gain of dropping α_2 comes entirely from those same priors, so the same widening surrenders the gain. A single knob – a common factor s on all seven exploration widths of Table 1 – therefore slides the fit between two ends: at $s = 1$ the full gain and almost no absorption, with flat priors the full absorption and no gain. The scan locates where between the two ends each layout and cut sits.

The scan. The fit is the one of the absorption test – the fit without α_2, which carries α_0 but no α_2: the α_0-only setup of Section 3 – fitted to the truth $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ in every group, with all seven σ_v prior widths multiplied by $s \in \{1, 1.5, 2, 3, 5, 10\}$ and flat. Each s is scored by $\delta_{\rm max}$, the largest of the (w_0, w_a) shift in its ellipse and the Ω_m and A_s shifts in their own σ, and by $\mathrm{FoM}_{\alpha_0\,\mathrm{only}}/\mathrm{FoM}$, that fit's $\mathrm{FoM}(w_0, w_a)$ over the reference model's at $s = 1$. The passing s is the smallest s with every shift below $\sigma/3$ – a bias under a third of the quoted error bar, small enough not to move any conclusion – interpolated linearly in $\log s$ between the scanned values. Figure 3 draws the scan; every cut, the per-shift values and the one-sample variants below are in that folder's `data.json`.

At $k_{\rm max} = 0.075\,h/\mathrm{Mpc}$ the exploration priors almost suffice. $\Delta d_{\rm com} = 62$ Mpc passes at $s = 1.35$, keeping 2.26 times the reference model's FoM; $\Delta z = 0.02$ passes at $s = 1$, keeping 2.74. $\Delta z = 0.05$ passes at $s = 1$ at every cut up to 0.15 (FoM ratios 1.09–1.17) and needs $s \approx 2.9$–3.0 at 0.175–0.2, where the ratio is 0.99–1.01; $\Delta z = 0.2$ passes at $s = 1$ everywhere (1.03–1.19).

At 0.1–0.2 on the thin layouts the two options converge. There the passing factor is $s = 3.56$–3.82 on $\Delta d_{\rm com} = 62$ Mpc and 4.93–6.29 on $\Delta z = 0.02$: ELG2's 30 km/s prior

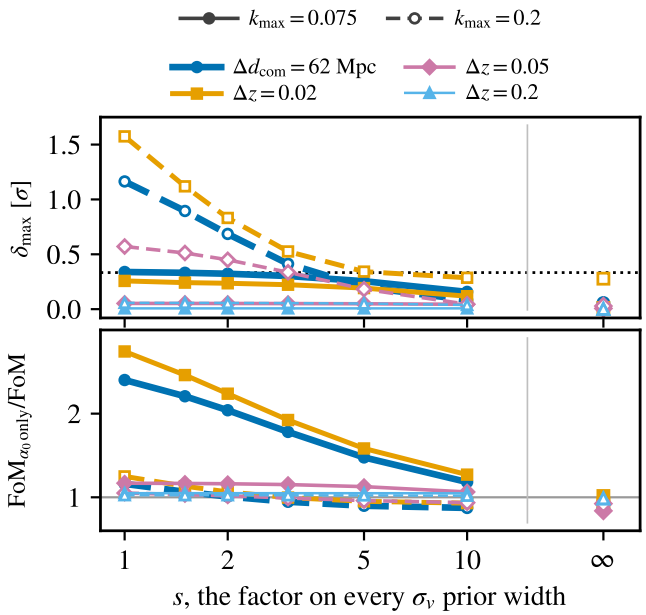


Figure 3: The σ_v prior-width scan: the fit without α_2 (the reference model with α_2 and its tilts fixed at zero, so α_0 is its only counterterm) fitted to a truth with $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ in every group, against s, the common factor on all seven σ_v prior widths (log axis; the flat-σ_v points sit at the right edge, past the gap), per layout (color, line width and marker; legend) at $k_{max} = 0.075\,h/\mathrm{Mpc}$ (solid, filled markers) and 0.2 (dashed, open markers). Top panel: δ_{max}, the largest of the (w_0, w_a) shift in its ellipse and the Ω_m and A_s shifts in their own σ; dotted: $\sigma/3$. Bottom panel: FoM$_{\alpha_0\,\mathrm{only}}$/FoM, that fit's FoM$(w_0, w_a)$ over the reference model's (α_2 free, σ_v at $s = 1$); gray: one. The paper model is this fit at $s = 6$. k_{max} quoted at $z = 0$; each bin is cut at $k_{max}/D(\bar{z}_i)$.

becomes about 110–190 km/s, a third to two thirds of its fiducial dispersion. At those widths the FoM ratio is 0.93–1.16: the widened α_0-only fit and the reference model constrain dark energy essentially equally, and the gain that motivated dropping α_2 is gone.

One sample's dispersions alone do not absorb it. The truth holds α_2 in every group, so the mimicking must too. Widening only the three LRG dispersions fails wherever $s = 1$ does not already pass: on $\Delta d_{com} = 62$ Mpc at 0.2, δ_{max} falls only from 1.16 to 1.13 even with the LRG priors flat. Widening only the two ELG dispersions passes on $\Delta d_{com} = 62$ Mpc at 0.075 ($s = 1.73$, ratio 2.33) and on $\Delta z = 0.05$ at 0.175–0.2, but fails on the thin layouts at 0.1 and above. Absorbing α_2 needs all the dispersions widened together.

The adopted width. The paper model takes $s = 6$ on every layout: the fit without α_2 with all seven widths of Table 1 multiplied by six, written into the paper yamls. Evaluated directly at $s = 6$, the $\alpha_2 = \pm 12.5\,h^{-2}\,\mathrm{Mpc}^2$ truth moves no score by more than 0.327σ at any layout or cut, so $s = 6$ passes the $\sigma/3$ criterion everywhere. The largest interpolated passing threshold of the scan is 6.3, on $\Delta z = 0.02$ at $k_{max} = 0.1\,h/\mathrm{Mpc}$, between the scanned $s = 5$ and 10 –

above the adopted 6 – and the 0.327σ sits on exactly that point, so the margin there is small. The prior-width truth, α_2 at one prior width $0.5/k_{\rm max}^2$ in every group, is a stress test rather than a DESI-size estimate, and the widened dispersions do not absorb it at the smaller cuts: it moves A_s by 1.6σ at 0.075 on $\Delta d_{\rm com} = 62$ Mpc and (w_0, w_a) by 1.3σ at 0.1 on $\Delta z = 0.02$, stays above $\sigma/3$ up to 0.125 on $\Delta d_{\rm com} = 62$ Mpc, up to 0.175 on $\Delta z = 0.02$ and at 0.075 and 0.125 on $\Delta z = 0.05$ (0.34σ at both), and moves $\Delta z = 0.2$ by at most 0.07σ.

The adopted width does not inflate the figure of merit. The width was chosen so that dropping α_2 does not buy constraining power from the σ_v priors. The check is the ratio of the paper model's FoM(w_0, w_a) to the reference model's (both on the same paper-fiducial cuts):

$k_{\rm max}$ [h/Mpc]	0.05	0.075	0.1	0.125	0.15	0.175	0.2
$\Delta d_{\rm com} = 62$ Mpc	1.52	1.39	1.02	0.92	0.91	0.90	0.89
$\Delta z = 0.02$	–	1.49	1.04	1.00	0.97	0.95	0.94
$\Delta z = 0.05$	1.15	1.11	1.06	1.02	1.00	0.97	0.95
$\Delta z = 0.2$	1.07	1.05	1.06	1.19	1.07	1.03	1.02

On the thin layouts the ratio is at most 1.04 at $k_{\rm max} = 0.1\,h$/Mpc and below one from 0.125 on, against the factor 1.7–2.1 at 0.1 of the fixed fit at $s = 1$; at 0.075 it is still 1.39–1.49, the remnant of that gain which the widening has not surrendered at so small a cut. The wide layouts are the exception: $\Delta z = 0.05$ keeps a ratio of 1.06 at 0.1 and $\Delta z = 0.2$ reaches 1.19 at 0.125, equal there to the $s = 1$ fixed-fit ratios of Section 5 – on those layouts the data barely constrain σ_v, so the widening changes nothing and the modest gain of removing the α_2 parameters remains. FoM$(\Omega_m, 10^9 A_s)$ behaves the same way: 1.45 and 1.44 at 0.075 on the thin layouts, at most 1.05 at 0.1 and 0.90–0.96 from 0.125 on, with the same wide-layout exceptions (up to 1.09 on $\Delta z = 0.05$ at 0.1 and 1.20 on $\Delta z = 0.2$ at 0.125).

7 The pair list at the edges of the σ_v prior

The rule. The data vector measures a fixed list of cross spectra, chosen by the Fingers-of-God overlap rule (`scripts/make_desi_pairs.py`; `notes/cross_correlations` derives it). With the normalized overlap of two bins' blurred $n(z)$,

$$O_{ij} \;=\; \frac{\int dz\, n_i(z)\, n_j(z)}{\sqrt{\int dz\, n_i^2(z)\, \int dz\, n_j^2(z)}}, \tag{7}$$

1 for two identical shapes, 0 for disjoint ones and, at high ℓ, close to the correlation coefficient of the two bins' spectra, the list holds DESI's measured pairs plus every pair with $O_{ij} \geq 10^{-6}$ at the fiducial σ_v. The list and the covariance are built once, at the fiducial dispersions; a sampled σ_v changes every kernel but never the list.

Raising σ_v pushes unlisted pairs toward the threshold. A larger σ_v widens every kernel, so every overlap grows – the listed ones, and those of the pairs outside the list, bins of one sample two apart, whose blurred kernels are separated by a full bin. Figure 4 follows the largest unlisted overlap as every dispersion group moves to $\sigma_v^{\rm fid} + k\, s\, \Delta\sigma_v$, with k the offset in prior widths and s the width factor of Section 6. At the fiducial no unlisted kernels come

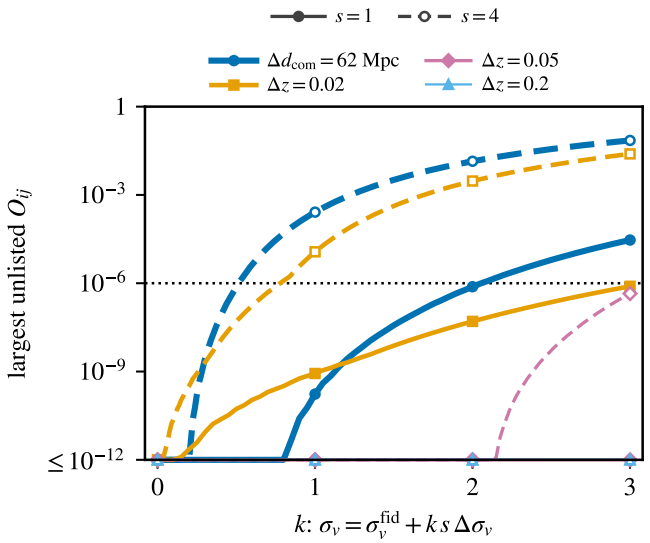


Figure 4: How far the pair rule holds away from the fiducial dispersions: the largest overlap O_{ij} of Eq. (7) over the pairs *outside* each layout's pair list, against the offset k of every σ_v group from its fiducial, in units of its exploration prior width times the width factor s ($\sigma_v = \sigma_v^{\rm fid} + k\,s\,\Delta\sigma_v$), per layout (color, line width and marker; legend). Solid, filled markers: $s = 1$, the exploration priors; dashed, open: $s = 4$, the order of the widths that let σ_v absorb a DESI-size α_2 (the paper model's $s = 6$ is quoted in the text); markers at $k = 0$, 1, 2, 3. Dotted: the rule's threshold $O_{ij} = 10^{-6}$, above which a pair belongs in the data vector. Overlaps below the axis floor, zero included (kernels that do not meet), are drawn on the floor.

near the threshold: the largest unlisted overlap is zero on three layouts and 1.5×10^{-14} on $\Delta z = 0.02$. At $k = 1$, 2 and 3 with $s = 1$ it reaches 1.7×10^{-10}, 7.6×10^{-7} and 2.9×10^{-5} on $\Delta d_{\rm com} = 62$ Mpc (a pair of BGS bins two apart), and 8.6×10^{-10}, 5.1×10^{-8} and 7.9×10^{-7} on $\Delta z = 0.02$ (ELG2, then LRG3 bins two apart), while $\Delta z = 0.05$ and 0.2 stay at zero. The threshold is crossed at $k = 2.06$ on $\Delta d_{\rm com} = 62$ Mpc, 3.11 on $\Delta z = 0.02$, 12.5 on $\Delta z = 0.05$ and nowhere on $\Delta z = 0.2$.

The reference model stays inside the list. At the $+2\sigma$ edge of the exploration priors the largest unlisted overlap is 7.6×10^{-7} on $\Delta d_{\rm com} = 62$ Mpc – a factor 1.3 below the threshold – and 5.1×10^{-8} on $\Delta z = 0.02$; at +2 posterior widths of the reference model it is the same (at 0.075, where the posterior equals the prior) or smaller (at 0.2). A chain over the exploration priors therefore keeps every pair it needs in the list; $\Delta d_{\rm com} = 62$ Mpc crosses only just beyond, at +2.1 prior widths.

The paper model's widened priors do not stay inside. With every width multiplied by $s = 6$, the crossing at $k\,s = 2.06$ and 3.11 prior widths sits at $k = 0.34$ and 0.52 of the *widened* width on $\Delta d_{\rm com} = 62$ Mpc and $\Delta z = 0.02$: well inside one sigma of the adopted prior. At one widened width the largest unlisted overlap is already 3.3×10^{-3} and 4.2×10^{-4}, and at the $+2\sigma$ edge it reaches 7.1×10^{-2} and 2.5×10^{-2} – four to five orders of magnitude

above the threshold, always a pair of BGS0 bins two apart. $\Delta z = 0.05$ crosses at 2.1 widened widths and $\Delta z = 0.2$ never. A chain with the paper priors that explores high BGS0 dispersions would therefore need cross spectra the data vector does not carry. What those lists measure is forecast in `notes/cross_correlations` (its σ_v axis, with the gain attributed to the free bias levels and slopes); the analysis keeps the current list, and those larger lists are a comparison against it.

What α_2 asks of σ_v. Only a *negative* α_2 drives the dispersions up: a positive α_2 looks like a smaller σ_v (Section 5), so its mirror image raises them. In the linear response of the fit without α_2, one unit of α_2 moves the most responsive dispersion by up to 2.2 km/s with the exploration priors on (the pull grows with the cut; at 0.075 it is 0.01–0.3 km/s across the three thinner layouts) and by 5.5–7.2 km/s with flat priors at cuts of 0.075 and above (on every layout but $\Delta z = 0.2$). The $|\alpha_2|$ at which an unlisted pair first reaches 10^{-6} is, with flat σ_v: 16–19 $h^{-2}\,\mathrm{Mpc}^2$ on $\Delta z = 0.02$ – 1.3–1.5 times the DESI prior width $12.5\,h^{-2}\,\mathrm{Mpc}^2$ – through a pair of ELG2 bins two apart; 53–59 on $\Delta d_{\mathrm{com}} = 62$ Mpc (ELG1); 104–133 on $\Delta z = 0.05$; and 400–466 on $\Delta z = 0.2$. With the exploration priors it is at least $69\,h^{-2}\,\mathrm{Mpc}^2$ at $k_{\mathrm{max}} = 0.2\,h/\mathrm{Mpc}$ and at least $1170\,h^{-2}\,\mathrm{Mpc}^2$ at 0.075.

The listed pairs' own overlaps. The complementary test – keeping only the listed pairs whose overlap reaches a floor O_{min}, DESI-measured pairs included – drops, for every floor up to 10^{-3}, one pair: the BGS–LRG1 boundary pair of $\Delta d_{\mathrm{com}} = 62$ Mpc ($O = 3.4 \times 10^{-7}$), at a cost of 0.1% of FoM(w_0, w_a) at $k_{\mathrm{max}} = 0.2\,h/\mathrm{Mpc}$ and nothing elsewhere, so none of this note's conclusions rests on a low-overlap pair. The study and its limits are in `notes/cross_correlations`, section "How much do the low-overlap pairs carry?"; the same scan of the paper model feeds the pair-threshold figure of `notes/bias_model_summary`.

8 What this means

At the template level, α_2 and the sampled σ_v carry the same $k_\parallel^2$ information: in that sense the model double counts one physical effect, and the α_2 group values add almost no template direction the dispersions do not already span. Only the α_2 tilts are new.

In the posterior, the Gaussian σ_v priors break the degeneracy. At the exploration widths (10–40% of the fiducial dispersions) the dispersions cannot wander along the degenerate direction, so the joint fit stays well conditioned at every cut, with the largest α_2–σ_v correlation reaching 0.9 only at the largest cuts on the thin layouts. Keeping both parameters then has a definite division of labor: α_2, with its wide prior, takes the $k_\parallel^2$ information, and the dispersions fall back to their priors (0.60–1.00 of the prior width with α_2 free). The data then constrain nonlinear velocities only as far as the α_2 prior allows, and the σ_v priors contribute almost nothing to the FoM.

The paper model takes the other route: it drops α_2 and widens every σ_v prior by $s = 6$, the factor at which the dispersions absorb a DESI-size α_2 wherever the data can see one. The widening surrenders the gain that dropping α_2 at the exploration widths would buy: from $k_{\mathrm{max}} = 0.1\,h/\mathrm{Mpc}$ on the thin layouts the paper model's FoM is within 4% of the reference model's or below it, while at 0.075 it keeps a factor 1.4–1.5 and the wide layouts keep their modest α_2-fixed gains (up to 1.06 on $\Delta z = 0.05$ at 0.1 and 1.19 on $\Delta z = 0.2$ at 0.125). In

exchange the model no longer risks the absorption shift: the $\alpha_2 = \pm 12.5\,h^{-2}\,\mathrm{Mpc}^2$ truth moves no score by more than 0.327σ anywhere – a linear-response estimate of MCMC safety, not a test the Fisher-only paper performs on data – though a truth at one α_2 prior width, a stress test, still shifts the small cuts by up to 1.6σ.

Two costs of the widened priors are measured here: the gain-free FoM now rests on priors six times wider than the measurement-based exploration widths – how defensible they are stays open (appendix "Open issues") – and the pair list is no longer complete at the widened prior's edge (Section 7); the lists selected there are forecast in `notes/cross_correlations` as a comparison against the current list, which the analysis keeps.

A Method and checks

Machinery. Everything runs in section 11 of `EXAMPLE_EVALUATE_FISHER_MODEL.ipynb`: the final-tier derivative caches of that notebook (five-point stencils; the α_2 columns are exact one-difference templates, since the term is linear in its amplitude to 2×10^{-10}), the Gaussian band-power covariance restricted to the bands each cut keeps, and the notebook's Fisher systems $F = \mathsf{D}^\mathsf{T}\mathsf{C}^{-1}\mathsf{D} + F_{\mathrm{priors}}$, with D the derivative matrix and F_{priors} the Gaussian priors of the fit at hand (the pessimistic linear-bias prior, the one-loop offset and counterterm priors, and the σ_v priors of Table 1 at the width the variant uses). Posterior correlations are read off F^{-1}; the fixed-α_2, flat-σ_v and widened-σ_v variants remove the α_2 columns, replace the σ_v prior rows or rescale them, and rebuild F.

The absorption test is a linear response. The shift of the fit without α_2 under an α_2 truth is the Fisher linear response $\delta p = F^{-1}\mathsf{D}^\mathsf{T}\mathsf{C}^{-1}\,\delta m$, with δm the truth's change of the data vector, scored in the fit's marginalized (w_0, w_a) ellipse (σ for one parameter) – not a refit. Shifts of one to two ellipse units extrapolate that linear response accordingly.

Checks. The analysis reproduces, to the printed precision (0.0 relative difference), the reference model's FoM(w_0, w_a) and FoM$(\Omega_m, 10^9 A_s)$ of the notebook's `fom_kmax_models` figure data at every cut, and the paper model's FoMs of its section 9 (the summary model). The $+12.5$ and $-12.5\,h^{-2}\,\mathrm{Mpc}^2$ truths shift the fit without α_2 equally, as the linear response requires (difference 0); the $s = 1$ and flat points of the width scan reproduce the absorption test's shifts to 2×10^{-9}, and the scan's fit at $s = 1$ reproduces the fixed-α_2 fit exactly. The pair-overlap cell reimplements the pair rule's blur on sliced kernels and agrees with `make_desi_pairs.overlap` to 9×10^{-16}. The σ_v-template fits of the $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ truth satisfy the linear-algebra identities of Eq. (6) by construction; the notebook's own self-checks on the α_2 columns (the data vector returns bitwise after the template evaluation; forward and backward differences agree) are listed with that notebook.

B Open issues

- **The basis of the σ_v priors.** The exploration widths are Gaussians five times as wide as the median per-bin error bar of the source forecast's measurement on simulated galaxies, truncated at zero (`notes/cross_correlations`); the paper widths are six times the exploration widths, a modeling choice justified by the absorption scan rather than by a

measurement. How defensible the widened widths are for the real samples, and whether a Gaussian blur is the right shape for the small-scale velocity distribution, are open.

- **No redshift-space one-loop terms.** The sampled σ_v and α_2 are the model's only handles on nonlinear velocities; a velocity effect beyond a $k_\parallel^2$ correction is outside both, and the paper model keeps only σ_v.

- **Linear response.** The largest absorption-test shifts (1.5–2.5 ellipse units) are linear extrapolations; a refit could move them.

- **The pair list at the σ_v prior edge.** The paper model's widened priors cross the pair rule's threshold well inside one sigma on the thin layouts (Section 7). The lists selected at those dispersions are forecast in `notes/cross_correlations` (its σ_v axis), with spectra, covariances and derivative caches built on each layout's largest list: the added pairs raise the figures of merit by separating each group's free bias level and slope from the growth history, not through line-of-sight information of their own. The analysis keeps the current list; those forecast lists are a comparison against it.

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